

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                             |                                                                       |
|-----------------------------|-----------------------------------------------------------------------|
| <b>Product Name</b>         | <u>Phosphorus pentasulfide</u>                                        |
| <b>CAS No</b>               | 1314-80-3                                                             |
| <b>Synonyms</b>             | Sulfur phosphide; Thiophosphoric anhydride; Diphosphorus pentasulfide |
| <b>Molecular Formula</b>    | P <sub>4</sub> S <sub>10</sub>                                        |
| <b>Molecular Weight</b>     | 444.48                                                                |
| <b>Recommended Use</b>      | Laboratory chemicals.                                                 |
| <b>Uses advised against</b> | No Information available                                              |

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>63005</b>                                                                                |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

**HSNO Approval Number**     **HSR003063**

### GHS Classification

#### Physical hazards

|                                                                        |                       |
|------------------------------------------------------------------------|-----------------------|
| Flammable solids                                                       | Category 1            |
| Substances/mixtures which, in contact with water, emit flammable gases | Category 1 Category 2 |

#### Health hazards

|                                             |            |
|---------------------------------------------|------------|
| Acute Oral Toxicity                         | Category 4 |
| Acute Inhalation Toxicity - Dusts and Mists | Category 1 |

#### Environmental hazards

|                          |            |
|--------------------------|------------|
| Acute aquatic toxicity   | Category 1 |
| Chronic aquatic toxicity | Category 1 |

**Label Elements****Signal Word****Danger****Hazard Statements**

H228 - Flammable solid  
H260 - In contact with water releases flammable gases which may ignite spontaneously  
H302 - Harmful if swallowed  
H261 - In contact with water releases flammable gases  
H330 - Fatal if inhaled  
H410 - Very toxic to aquatic life with long lasting effects

**Precautionary Statements****Prevention**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P223 - Do not allow contact with water  
P231 + P232 - Handle and store contents under inert gas. Protect from moisture  
P240 - Ground and bond container and receiving equipment  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P273 - Avoid release to the environment

**Response**

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P330 - Rinse mouth  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P302 + P335 + P334 - IF ON SKIN: Brush off loose particles from skin. Immerse in cool water  
P391 - Collect spillage

**Storage**

P402 + P404 - Store in a dry place. Store in a closed container

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

Lachrymator (substance which increases the flow of tears) Stench  
May form explosible dust-air mixture if dispersed  
May form combustible dust concentrations in air  
Contact with water liberates toxic gas

## Section 3 - Composition and Information on Ingredients

| Component               | CAS No    | Weight % |
|-------------------------|-----------|----------|
| Phosphorus pentasulfide | 1314-80-3 | >95      |

## Section 4 - First Aid Measures

**Description of first aid measures**

|                                            |                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                                                                                                     |
| <b>New Zealand Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                                                                                                                                                                                                                                                                                            |
| <b>Inhalation</b>                          | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration. Call a physician or poison control center immediately. |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                     |
| <b>Skin Contact</b>                        | Immediate medical attention is required. Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.                                                                                                                                                                                                                                        |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately. Immediate medical attention is required. Remove from exposure, lie down. Clean mouth with water and drink afterwards plenty of water. Never give anything by mouth to an unconscious person.                                                                                                           |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                                                                      |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                                                                  |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. May cause pulmonary edema: Symptoms may be delayed                                                                                                                                                                                                                                                                                                           |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                                                                                                                |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water mist may be used to cool closed containers. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

### Extinguishing media which must not be used for safety reasons

Contact with water liberates toxic gas.

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Contact with water liberates toxic gas. May form explosive mixtures with air. Produce flammable gases on contact with water. Thermal decomposition can lead to release of irritating gases and vapors. In the event of fire and/or explosion do not breathe fumes. Contact with water liberates toxic gas. Do not allow run-off from fire-fighting to enter drains or water courses. Fine dust dispersed in air may ignite.

### Hazardous Combustion Products

Hydrogen sulfide (H<sub>2</sub>S), Sulfur oxides, Oxides of phosphorus.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

#### Emergency procedures

Use personal protective equipment as required. Avoid dust formation. Remove all sources of ignition. Take precautionary measures against static discharges. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. Do not expose spill to water.

**Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling****Advice on safe handling**

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. Keep away from open flames, hot surfaces and sources of ignition. Do not taste or swallow. Do not allow contact with water.

**Hygiene Measures**

Keep away from food, drink and animal feeding stuffs. When using do not eat, drink or smoke. Contaminated work clothing should not be allowed out of the workplace. Provide regular cleaning of equipment, work area and clothing. Avoid contact with skin, eyes or clothing. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wear suitable gloves and eye/face protection.

**Conditions for Safe Storage, Including any Incompatibilities****Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place. Flammables area. Keep away from heat, sparks and flame. Keep in properly labeled containers. Keep from any possible contact with water. Keep away from water or moist air.

**Incompatible Materials**

Strong oxidizing agents.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Control parameters****Exposure limits**

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

| Component               | New Zealand WEL          | Australia                                             | ACGIH TLV                                             | The United Kingdom                                                |
|-------------------------|--------------------------|-------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------|
| Phosphorus pentasulfide | TWA: 1 mg/m <sup>3</sup> | STEL: 3 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup> | STEL: 2 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Appropriate engineering controls**Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Individual protection measures, such as personal protective equipment

|                        |                                                                                                                                               |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Protection</b>  | Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications) |
| <b>Hand Protection</b> | Protective gloves                                                                                                                             |

| Glove material                                 | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Natural rubber, Nitrile rubber, Neoprene, PVC. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                 |                                                                                                                                                                                                                                                                                                                             |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure Impervious clothing Chemical resistant apron Boots Impervious gloves                                                                                                                                                                               |
| <b>Respiratory Protection</b>   | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices |
| <b>Recommended Filter type:</b> | Particulates filter conforming to EN 143 (or AUS/NZ equivalent)                                                                                                                                                                                                                                                             |
| <b>Recommended half mask:-</b>  | Particle filtering: EN149:2001 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                                                                                         |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hygiene Measures</b> | Keep away from food, drink and animal feeding stuffs. When using do not eat, drink or smoke. Contaminated work clothing should not be allowed out of the workplace. Provide regular cleaning of equipment, work area and clothing. Avoid contact with skin, eyes or clothing. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wear suitable gloves and eye/face protection. |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                        |                                                                                                                                                                                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained. |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Section 9 - Physical and Chemical Properties

Information on basic physical and chemical properties

|                                 |                               |
|---------------------------------|-------------------------------|
| <b>Physical State</b>           | Solid                         |
| <b>Appearance</b>               | Yellow                        |
| <b>Odor</b>                     | Stench                        |
| <b>Odor Threshold</b>           | No data available             |
| <b>pH</b>                       | 1                             |
| <b>Melting Point/Range</b>      | 286 - 290 °C / 546.8 - 554 °F |
| <b>Softening Point</b>          | No data available             |
| <b>Boiling Point/Range</b>      | 514 °C / 957.2 °F             |
| <b>Flammability (liquid)</b>    | Not applicable                |
| <b>Flammability (solid,gas)</b> | No information available      |
| <b>Explosion Limits</b>         | No data available             |

|                                         |                                                                    |                                   |
|-----------------------------------------|--------------------------------------------------------------------|-----------------------------------|
| Flash Point                             | > 100 °C / > 212 °F                                                | Method - No information available |
| Autoignition Temperature                | 282 - °C / 539.6 - °F                                              |                                   |
| Decomposition Temperature               | No data available                                                  |                                   |
| Viscosity                               | Not applicable                                                     | Solid                             |
| Water Solubility                        | reacts forming H <sub>3</sub> PO <sub>4</sub> and H <sub>2</sub> S |                                   |
| Solubility in other solvents            | No information available                                           |                                   |
| Partition Coefficient (n-octanol/water) |                                                                    |                                   |
| Vapor Pressure                          | 1 mmHg @ 300 °C                                                    |                                   |
| Density / Specific Gravity              | 2.080                                                              |                                   |
| Bulk Density                            | No data available                                                  |                                   |
| Vapor Density                           | Not applicable                                                     | Solid                             |
| Particle characteristics                | No data available                                                  |                                   |

#### Other information

|                                                                        |                                                                                    |
|------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Molecular Formula                                                      | P <sub>4</sub> S <sub>10</sub>                                                     |
| Molecular Weight                                                       | 444.48                                                                             |
| Flammable solids                                                       | Burning rate or burning time = > 2.2 mm/s or < 45 secs<br>Wetted zone passed - Yes |
| Substances/mixtures which, in contact with water, emit flammable gases | Emitted gas ignites spontaneously<br>Gas(es) = Hydrogen sulfide (H <sub>2</sub> S) |
| Evaporation Rate                                                       | Not applicable - Solid                                                             |

## Section 10 - Stability and Reactivity

|                                  |                                                                                                                                                                                                                                       |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reactivity                       | Yes                                                                                                                                                                                                                                   |
| Stability                        | Moisture sensitive. Reacts violently with water.                                                                                                                                                                                      |
| Sensitivity to Mechanical Impact | No information available                                                                                                                                                                                                              |
| Sensitivity to Static Discharge  | No information available                                                                                                                                                                                                              |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                                                                                                                              |
| Hazardous Reactions              | Contact with water liberates toxic gas.                                                                                                                                                                                               |
| Conditions to Avoid              | Incompatible products, Excess heat, Keep away from open flames, hot surfaces and sources of ignition, Avoid dust formation, Exposure to moist air or water, Exposure to air or moisture over prolonged periods, Exposure to moisture. |
| Incompatible Materials           | Strong oxidizing agents.                                                                                                                                                                                                              |
| Hazardous Decomposition Products | Hydrogen sulfide (H <sub>2</sub> S). Sulfur oxides. Oxides of phosphorus.                                                                                                                                                             |

## Section 11 - Toxicological Information

#### Acute Effects

#### Information on likely routes of exposure

#### Product Information

|            |                                                                                                                                                                                      |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation | Harmful by inhalation. May cause irritation of respiratory tract.                                                                                                                    |
| Eyes       | May cause irritation. Causes burns. Corrosive to the eyes and may cause severe damage including blindness. Risk of serious damage to eyes.                                           |
| Skin       | Harmful in contact with skin. May cause irritation. Causes burns.                                                                                                                    |
| Ingestion  | Harmful if swallowed. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea. Ingestion causes burns of the upper digestive and respiratory tracts. Can burn |

mouth, throat, and stomach.

#### Numerical measures of toxicity

**(a) acute toxicity;**

**Oral**

Category 4

**Dermal**

Based on available data, the classification criteria are not met

**Inhalation**

Category 4

| Component               | LD50 Oral                              | LD50 Dermal           | LC50 Inhalation |
|-------------------------|----------------------------------------|-----------------------|-----------------|
| Phosphorus pentasulfide | 791 mg/kg ( Rat )<br>389 mg/kg ( Rat ) | 3160 mg/kg ( Rabbit ) |                 |

**(b) skin corrosion/irritation;**

Based on available data, the classification criteria are not met

**(c) serious eye damage/irritation;**

Based on available data, the classification criteria are not met

**(d) respiratory or skin sensitization;**

**Respiratory**

Based on available data, the classification criteria are not met

**Skin**

Based on available data, the classification criteria are not met

**(e) germ cell mutagenicity;**

Based on available data, the classification criteria are not met

**(f) carcinogenicity;**

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;**

Based on available data, the classification criteria are not met

**(h) STOT-single exposure;**

Based on available data, the classification criteria are not met

**(i) STOT-repeated exposure;**

Based on available data, the classification criteria are not met

**Target Organs**

None known.

**(j) aspiration hazard;**

Not applicable

Solid

**Symptoms / effects, both acute and delayed**

May cause pulmonary edema. Symptoms may be delayed.

## Section 12 - Ecological Information

#### Ecotoxicity

**Aquatic ecotoxicity**

Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component               | Freshwater Fish | Water Flea             | Freshwater Algae | Microtox |
|-------------------------|-----------------|------------------------|------------------|----------|
| Phosphorus pentasulfide |                 | EC50 = 0.16 mg/L (48h) |                  |          |

**Terrestrial ecotoxicity**

There is no data for this product

|                                              |                                                                                                                                                                                                                         |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Persistence and Degradability</b>         | Not readily biodegradable                                                                                                                                                                                               |
| <b>Persistence</b>                           | Persistence is unlikely, based on information available, Soluble in water.                                                                                                                                              |
| <b>Degradability</b>                         | Not relevant for inorganic substances, Reacts with water.                                                                                                                                                               |
| <b>Degradation in sewage treatment plant</b> | Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.                                                                                                         |
| <b>Bioaccumulative Potential</b>             | Product does not bioaccumulate due to reaction with water Bioaccumulation is unlikely                                                                                                                                   |
| <b>Mobility</b>                              | Reacts with water. The product is water soluble, and may spread in water systems. Is not likely mobile in the environment. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils |
| <b><u>Other adverse effects</u></b>          |                                                                                                                                                                                                                         |
| <b>Endocrine Disruptor Information</b>       | This product does not contain any known or suspected endocrine disruptors                                                                                                                                               |
| <b>Persistent Organic Pollutant</b>          | This product does not contain any known or suspected substance                                                                                                                                                          |
| <b>Ozone Depletion Potential</b>             | This product does not contain any known or suspected substance                                                                                                                                                          |

## Section 13 - Disposal Considerations

### Waste treatment methods

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.                                                                                                                                                      |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                                                                                                                                                                                  |
| <b>Other Information</b>                   | Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Solutions with low pH-value must be neutralized before discharge. Do not let this chemical enter the environment. Do not empty into drains. |

## Section 14 - Transport Information

| Component                                    | Hazchem Code |
|----------------------------------------------|--------------|
| Phosphorus pentasulfide<br>1314-80-3 ( >95 ) | 4W           |

### NZS 5433:2020

|                                |                          |
|--------------------------------|--------------------------|
| <b>UN-No</b>                   | UN1340                   |
| <b>Proper Shipping Name</b>    | PHOSPHORUS PENTASULPHIDE |
| <b>Hazard Class</b>            | 4.3                      |
| <b>Subsidiary Hazard Class</b> | 4.1                      |
| <b>Packing Group</b>           | II                       |

### IATA



|                         |                          |
|-------------------------|--------------------------|
| UN-No                   | UN1340                   |
| Proper Shipping Name    | PHOSPHORUS PENTASULPHIDE |
| Hazard Class            | 4.3                      |
| Subsidiary Hazard Class | 4.1                      |
| Packing Group           | II                       |

**IMDG/IMO**

|                         |                          |
|-------------------------|--------------------------|
| UN-No                   | UN1340                   |
| Proper Shipping Name    | PHOSPHORUS PENTASULPHIDE |
| Hazard Class            | 4.3                      |
| Subsidiary Hazard Class | 4.1                      |
| Packing Group           | II                       |

|                       |                                                                                                          |
|-----------------------|----------------------------------------------------------------------------------------------------------|
| Environmental hazards | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO |
|-----------------------|----------------------------------------------------------------------------------------------------------|

|                                                                          |                                |
|--------------------------------------------------------------------------|--------------------------------|
| Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code | Not applicable, packaged goods |
|--------------------------------------------------------------------------|--------------------------------|

|                     |                                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------|
| Special Precautions | No special precautions required. Please refer to the applicable dangerous goods regulations for additional information. |
|---------------------|-------------------------------------------------------------------------------------------------------------------------|

|                        |            |
|------------------------|------------|
| Additional information | None known |
|------------------------|------------|

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                      |           |
|----------------------|-----------|
| HSNO Approval Number | HSR003063 |
|----------------------|-----------|

**National Regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

**International Regulations**

|                           |                                                                |
|---------------------------|----------------------------------------------------------------|
| Ozone Depletion Potential | This product does not contain any known or suspected substance |
|---------------------------|----------------------------------------------------------------|

|                              |                                                                |
|------------------------------|----------------------------------------------------------------|
| Persistent Organic Pollutant | This product does not contain any known or suspected substance |
|------------------------------|----------------------------------------------------------------|

|                            |                |
|----------------------------|----------------|
| Rotterdam Convention (PIC) | Not applicable |
|----------------------------|----------------|

|                                                  |                |
|--------------------------------------------------|----------------|
| Authorisation/Restrictions according to EU REACH | Not applicable |
|--------------------------------------------------|----------------|

**International Inventories**

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component               | CAS No    | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL     | IECSC | TCSI |
|-------------------------|-----------|-------|------|-----------|--------|-----|----------|-------|------|
| Phosphorus pentasulfide | 1314-80-3 | X     | X    | 215-242-4 | -      | -   | KE-12128 | X     | X    |

| Component               | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|-------------------------|-----------|------|-----------------------------------------------|-----|------|-------|------|------|
| Phosphorus pentasulfide | 1314-80-3 | X    | ACTIVE                                        | X   | -    | X     | X    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

### Legend

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**NZS 5433:2020** - Transport of Dangerous Goods on Land

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**AICS** - Australian Inventory of Chemical Substances

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists

**PNEC** - Predicted No Effect Concentration

**OECD** - Organisation for Economic Co-operation and Development

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** - Australian Code for the Transport of Dangerous Goods by Road and Rail

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

### Training Advice

Chemical incident response training.

### Revision Date

10-Mar-2023

### Revision Summary

Not applicable

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

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**End of Safety Data Sheet**